

L9 P1 v0.8 Polish Notes — Substrate-Symplectic Cat 6 + Composite v0.4 Integration

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L9 P1 v0.8 Polish Notes

0. Goal

Per Casey Thursday PM L9: P1 v0.8 polish notes for incremental update from v0.7 SUBMISSION-FINAL (June 2 Casey authorization) absorbing Thursday substrate-mechanism advances.

1. v0.8 Polish Items

Item 1: Cat 6 Substrate-Symplectic Section Insert

Insert short subsection after v0.7 Section on substrate-K-types referencing $\mathrm{Sp}(2) \cong \mathrm{Spin}(5)$ accidental isomorphism and substrate-Hamiltonian flow at substrate-K-type spectrum.

Approximately 1 paragraph: “Substrate-K-types V_{λ_1, λ_2} on D_{IV}^5 inherit substrate-symplectic structure via $\mathrm{Spin}(5) \cong \mathrm{Sp}(2)$ accidental Lie-group isomorphism. Substrate-Hamiltonian flow on substrate-K-type spectrum is encoded by $\mathrm{Sp}(2)$ substrate-symplectic 2-form ω substrate-canonical antisymmetric. Substrate-time evolution $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$ at substrate-K-type substrate-Hamiltonian flow level.”

Item 2: Composite v0.4 Substrate-Mass-Mechanism Box

Insert short boxed result after substrate-Bergman matrix element discussion:

Substrate-mass-mechanism FORCING candidate at $\text{gen-2 } V_{(3/2, 1/2)}$:

$$\begin{aligned} m_{\mu} / m_e &= n_C \cdot (n_C / \text{rank})_3 + N_c^4 / 2^{N_c} \\ &= 5 \cdot 315/8 + 81/8 = 1656/8 = 207 \end{aligned}$$

precision 0.112% Tier 1/Tier 2 boundary vs observed ≈ 206.768

Term 1: substrate-Bergman + substrate-Pochhammer Schur II

Term 2: substrate-Casey-#14-codim-4 + substrate-color N_c

PARTIAL substrate-mechanism FORCING candidate per Cal #35 STANDING;
multi-week explicit FORCING derivation per Cal #189.

Item 3: Schur II Pochhammer Cross-Link

Substrate-Pochhammer $(n_C / \text{rank})_k = (5/2)_k$ substrate-natural form per $\text{Sym}^k V_{\text{fund}} \mathrm{Sp}(2)$ substrate-K-type representation at substrate-Bergman kernel exponent.

For substrate-spinor $V_{\lambda_1, \lambda_2} \text{Sym}^k$ $k = \lambda_1 + 1/2$ substrate-natural: - $\text{gen-1 } V_{(1/2, 1/2)}$: $k = 1, (5/2)I = 5/2$ - $\text{gen-2 } V_{(3/2, 1/2)}$: $k = 2$ or 3 (Sym^2 or Sym^3 — clarify in v0.8) - $\text{gen-3 } V_{(5/2, 1/2)}$: $k = 3$ or 5 (Sym^3 or Sym^5 — clarify in v0.8)

Item 4: Cal #36 STANDING Reference Update

Update Cal #36 STANDING reference per Thursday RATIFIED state (was candidate at v0.7).

2. v0.8 Polish Tier Status

- Item 1 (Cat 6 substrate-symplectic): substantive 1-paragraph insert; ready for v0.8 polish
- Item 2 (Composite v0.4 box): substantive boxed result; ready for v0.8 polish
- Item 3 (Schur II Pochhammer clarification): substantive cross-link; substantive multi-week clarification per Cal #189
- Item 4 (Cal #36 STANDING reference update): substantive small edit; ready for v0.8 polish

3. Closure v0.1

L9 P1 v0.8 polish notes v0.1: 4 polish items identified for v0.8 incremental update from v0.7 SUBMISSION-FINAL.

Items 1+2+4 ready for substantive v0.8 polish; Item 3 multi-week clarification per Cal #189.

Tier: L9 P1 v0.8 polish notes v0.1 framework; substantive substrate-mechanism advances absorbed for v0.8 incremental update.

— Lyra, Thu 2026-06-04 11:21 EDT. L9 P1 v0.8 polish notes v0.1: 4 polish items (Cat 6 substrate-symplectic + Composite v0.4 box + Schur II Pochhammer cross-link + Cal #36 STANDING reference update); substantive v0.8 polish ready for Items 1+2+4; Item 3 multi-week per Cal #189.